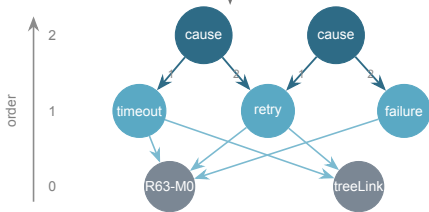


Representation: artifact $\rightarrow$ typed predicate DAG

raw session (BGL)

```
1117841440 R63-M0 cioid: timeout tree link  
1117841452 R63-M0 cioid: retry tree link  
1117841498 R63-M0 kernel panic
```

Tier-0 encoder (deterministic rules)



*shared sub-expressions are hash-consed
 $\Rightarrow$ the case is a DAG, not a bag; this is
what makes systematicity a graph property*

```
(cause (timeout R63-M0 treeLink) (retry R63-M0 treeLink)) case_id = BLAKE3(canonical) = 6c1f6a...
```